

Case Study: Telangana Heat Action Plan (HAP)

Institutional Frameworks, Urban Cooling Policies, and Climate Resilience Strategies

Policy Analysis Report

July 30, 2026

Abstract

This case study examines the development, structural mechanics, key innovations, and operational challenges of the Telangana State Heat Action Plan (HAP). As rising global temperatures intensify extreme heat events across Southern India, Telangana has emerged as a key innovator in sub-national climate adaptation. By integrating early warning systems, healthcare preparedness, sectoral labor protections, and structural interventions such as the *Telangana Cool Roofs Policy*, the state provides a valuable model for heat risk governance in developing economies.

1 Introduction & Context

Telangana, located in central-southern India, regularly experiences severe summer heatwaves between March and June. Temperatures frequently exceed 45°C in vulnerable interior districts such as Nalgonda, Ramagundam, and Khammam.

Following severe heatwave events across India in the mid-2010s, Telangana institutionalized a comprehensive **State Heat Action Plan (HAP)** in coordination with:

- **NDMA:** National Disaster Management Authority
- **IMD:** India Meteorological Department
- **TSDPS:** Telangana State Development Planning Society
- Technical and academic public health organizations

A key policy milestone occurred when Telangana declared heatwaves a **state-specific disaster**, allowing state authorities to allocate up to 10% of the State Disaster Response Fund (SDRF) directly toward heat relief and mitigation measures.

2 Institutional Framework & Core Pillars

The HAP coordinates across more than 20 state government departments. Its key functional mechanisms are illustrated below:

2.1 Early Warning Systems (EWS) & Public Communication

The system operates on hyper-local forecast dissemination:

- **Color-Coded Alert System:** Risk levels are classified into **Yellow** ($41^{\circ}\text{C} - 43^{\circ}\text{C}$), **Orange** ($43.1^{\circ}\text{C} - 44.9^{\circ}\text{C}$), and **Red Alerts** ($\geq 45^{\circ}\text{C}$).
- **Targeted Outreach:** Direct communication via SMS alerts, WhatsApp broadcast groups, local media, and field-level Accredited Social Health Activists (ASHA).

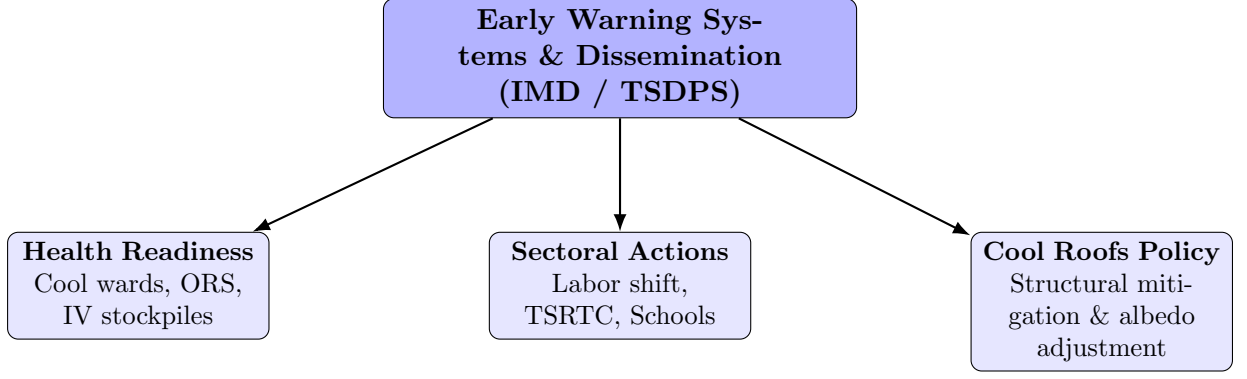


Figure 1: Organizational Structure of Telangana Heat Action Plan Interventions.

2.2 Healthcare System Preparedness

To minimize mortality from Heat-Related Illnesses (HRI):

- Dedicated *Cool Wards* equipped with climate control or cooling fans are reserved in primary healthcare centers (PHCs) and district hospitals.
- Strategic stockpiling of Oral Rehydration Salts (ORS), intravenous (IV) fluids, and ice packs.
- Mandatory daily surveillance and reporting of heatstroke admissions and mortalities.

2.3 Public Infrastructure and Labor Protection

Administrative directives adjust daily societal routines:

- Regulation of working hours for outdoor construction, agriculture, and manufacturing, enforcing mandatory rest breaks between 12:00 PM and 4:00 PM.
- Rescheduling of school hours to morning shifts during peak heat months.
- Provision of shaded waiting areas and drinking water stations (*Chalivendrams*) across public transit nodes managed by the Greater Hyderabad Municipal Corporation (GHMC) and Telangana State Road Transport Corporation (TSRTC).

3 Key Innovation: Telangana Cool Roofs Policy

To complement emergency response with long-term urban cooling, Telangana launched the **Telangana Cool Roofs Policy (2023–2028)**.

3.1 Technical Principle

The policy leverages High-Albedo materials (reflective paints, specialized tiles, or vegetated roofs) to modify the surface energy balance:

$$Q_{\text{absorbed}} = (1 - \alpha) \cdot I_{\text{solar}} \quad (1)$$

Where α represents the roof surface albedo and I_{solar} is the incoming solar irradiance. Increasing α reduces thermal energy transferred to the building interior, reducing indoor temperatures by 2°C to 5°C.

3.2 Scope and Targets

- **Mandatory Adoption:** Required for non-residential, government, and high-rise residential construction projects exceeding specified built-up area thresholds.
- **Low-Income Housing Integration:** Deployment across informal settlements and government housing schemes to provide thermal comfort to socio-economically vulnerable populations.

4 Phased Implementation Framework

The plan adopts a seasonal workflow divided into three explicit operational cycles:

Table 1: Seasonal Phased Implementation of the Telangana HAP

Phase	Timeline	Primary Operational Focus
Phase I: Pre-Heat	Jan – Mar	Interdepartmental coordination meetings, stocktaking of ORS/IV supplies, training medical personnel, conducting public awareness campaigns (IEC).
Phase II: Heat Event	Apr – Jun	Real-time EWS generation, monitoring daily HRI admissions, enforcing labor hour restrictions, ensuring emergency water supply.
Phase III: Post-Heat	Jul – Sep	Mortality data audits, interdepartmental performance evaluations, updating district-level vulnerability maps.

5 Critical Analysis and Outcomes

5.1 Key Successes

1. **Reduction in Acute Heat Mortality:** Structured healthcare protocols and public early warnings have significantly reduced direct heatstroke mortality compared to pre-2016 baseline figures.
2. **Institutionalizing Adaptation Finance:** Accessing the State Disaster Response Fund (SDRF) established financial predictability for emergency heat operations.
3. **Policy Integration:** Bridging emergency disaster management with long-term urban planning via the Cool Roofs initiative.

5.2 Challenges and Policy Gaps

- **Enforceability in Informal Sectors:** Labor advisories regarding outdoor work hours remain non-binding guidance, offering limited protection to gig workers, street vendors, and informal laborers.
- **Urban Heat Island (UHI) Amplification:** Rapid urban expansion and loss of green space continue to offset microclimate cooling efforts in urban corridors like Hyderabad.
- **Granular Data Collection:** Under-reporting of secondary heat-related morbidities (e.g., cardiovascular or renal aggravation caused by heat) remains an issue due to systemic diagnostic limitations.

6 Conclusion

The Telangana Heat Action Plan represents a progressive, multi-sectoral model for climate resilience in heat-vulnerable developing regions. While short-term emergency alerts and health-care interventions effectively reduce acute mortality, long-term success depends on structural cooling initiatives, legislative protections for informal outdoor labor, and sustainable urban spatial planning.